

双棱镜

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2025 年 11 月 26 日

x_1	x_2	$\Delta x = \frac{ x_1 - x_2 }{4}$
2.218 mm	3.944 mm	0.4315 mm

$x_{\text{狭缝}}$	$x_{\text{目镜}}$	$x_{\text{凸透镜}}$
146.85 cm	78.57 cm	102.50 cm

x'_1	x'_2
5.137 mm	5.517 mm

$$a = x_{\text{狭缝}} - x_{\text{凸透镜}} = 44.35 \text{ cm}$$

$$b = x_{\text{凸透镜}} - x_{\text{目镜}} = 23.93 \text{ cm}$$

$$D = x_{\text{狭缝}} - x_{\text{目镜}} - 15.2 \text{ cm} = 53.08 \text{ cm}$$

$$x'_1 = 5.137 \text{ mm}, x'_2 = 5.517 \text{ mm}$$

$$L' = x'_2 - x'_1 = 0.380 \text{ mm}$$

$$L = \frac{a}{b} L' = 0.7043 \text{ mm}$$

$$\lambda = L \frac{\Delta x}{d} = 573.9 \text{ nm}$$

不确定度的计算

由于我们只进行了一次测量我们只能计算 b 类不确定度.

$$u_a = u_a^B = \frac{0.1 \text{ cm}}{\sqrt{3}} = 5.773502691896 \times 10^{-2} \text{ m}$$

$$u_b = u_b^B = \frac{0.01 \text{ cm}}{\sqrt{3}} = 5.773502691896 \times 10^{-2} \text{ m}$$

$$u_{L'} = u_{L'}^B = \frac{0.01 \text{ mm}}{\sqrt{3}} = 5.773502691896 \times 10^{-4} \text{ m}$$

$$u_{\Delta x} = u_{\Delta x}^B = \frac{0.01 \text{ mm}}{\sqrt{3}} = 5.773502691896 \times 10^{-4} \text{ m}$$

$$U = \sqrt{\left(\frac{\partial \lambda}{\partial a}\right)^2 u_a^2 + \left(\frac{\partial \lambda}{\partial b}\right)^2 u_b^2 + \left(\frac{\partial \lambda}{\partial L'}\right)^2 u_{L'}^2 + \left(\frac{\partial \lambda}{\partial \Delta x}\right)^2 u_{\Delta x}^2}$$

$$\begin{aligned}\frac{\partial \lambda}{\partial a} &= \frac{1}{b} L' \frac{\Delta x}{d} = \frac{\lambda}{a} \\ \frac{\partial \lambda}{\partial b} &= -\frac{a}{b^2} L' \frac{\Delta x}{d} = -\frac{1}{b} \lambda \\ \frac{\partial \lambda}{\partial L'} &= \frac{a}{b} \frac{\Delta x}{d} = \frac{\lambda}{L'} \\ \frac{\partial \lambda}{\partial \Delta x} &= \frac{a}{b} L' \frac{1}{d} = \frac{\lambda}{\Delta x}\end{aligned}$$

$$U = \lambda \sqrt{\left(\frac{u_a}{a}\right)^2 + \left(-\frac{u_b}{b}\right)^2 + \left(\frac{u_a}{L'}\right)^2 + \left(\frac{u_{\Delta x}}{\Delta x}\right)^2 + \left(\frac{u_D}{\Delta D}\right)^2} = 24.5nm$$

则

$$\lambda = (573.9 \pm 24.5)nm$$